

Table 8.2 averages male and female mortality for simplicity and examines the probability that a given needs-to-wealth ratio is sustainable when market returns and life spans are random.

TABLE 8.2 What Is the Probability That Spending Is Sustainable?

Monte Carlo Simulation Results					
Retirement Age	Needs-to-Wealth Ratio				
	5.0%	6.0%	7.0%	8.0%	9.0%
55	71.3%	60.4%	50.1%	41.0%	33.3%
65	83.2%	74.7%	65.9%	57.3%	49.5%
75	93.9%	89.5%	84.2%	78.3%	72.3%

Simulation assumptions: Moderate mortality assumptions;
Equity: expected returns =7%, volatility = 20%
Source: Moshe Milevsky and the IFID Centre, 2008.

Your question at this point, of course, is how and where did I come up with these numbers? I'm glad you asked. I borrowed a technique developed by scientists over the last few decades to deal with complicated questions in nuclear physics. These days, it's used in everything from traffic control to designing better soap. It's called the method of Monte Carlo simulation. We used this method in earlier chapters; now let me explain how it works in more detail.

Together with some colleagues at the Individual Finance and Insurance Decisions (IFID) Centre in Toronto, Canada, we constructed a computer program that generates millions of different scenarios for the financial markets and human mortality. Figure 8.2 illustrates how three of these scenarios or paths might have looked. It is, if you like, the ultimate imagination machine. In one scenario you live to the ripe old age of 97; in another scenario you live to age 86. Some scenarios show the stock market booming for the next 20 years; others indicate a 10-year bear market.

As you might know, simulations of future market behavior have been employed quite successfully in corporate risk management, when a company wants to compute the probability of losing a specific amount of money over a particular time horizon. Government